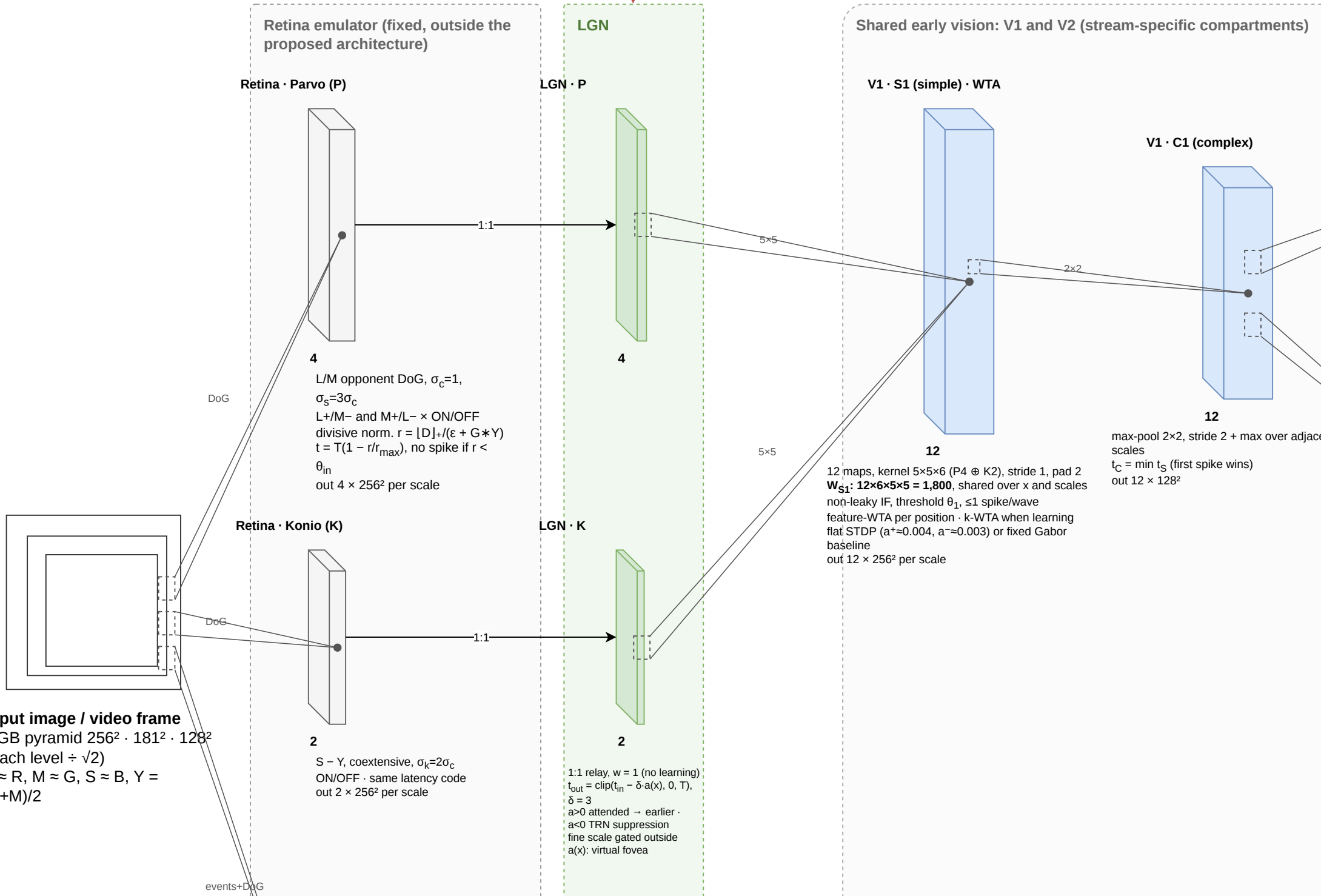
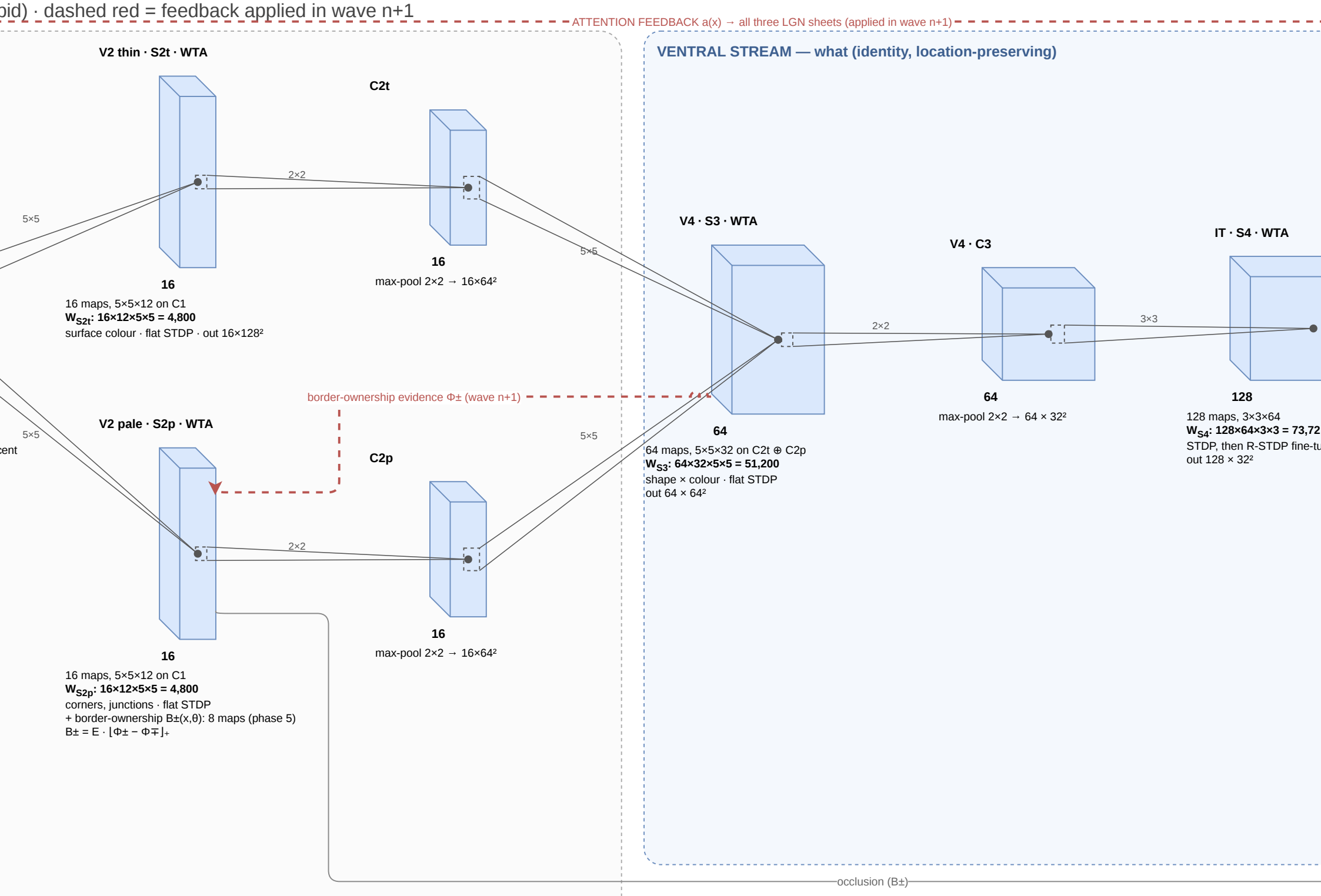
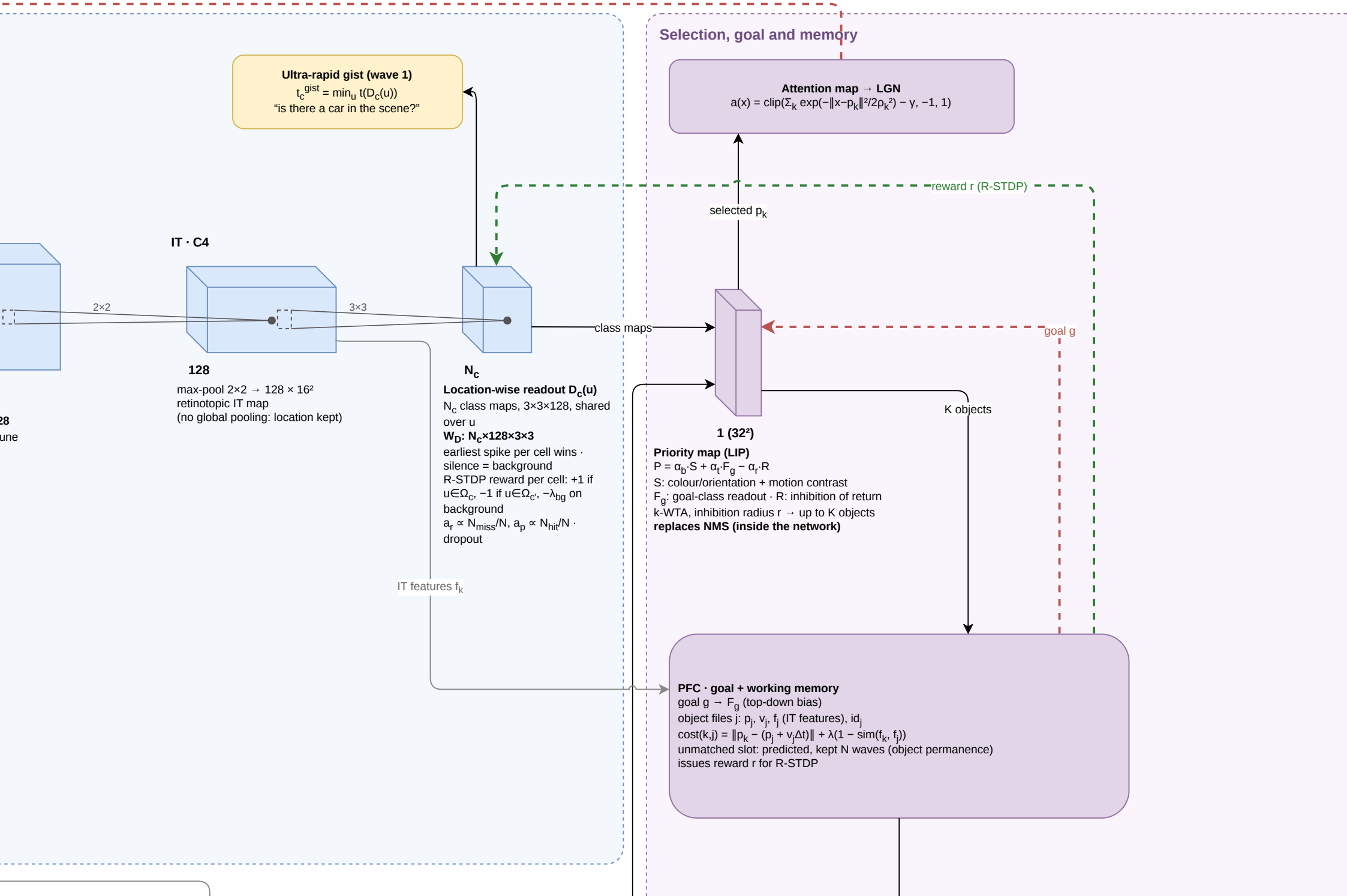
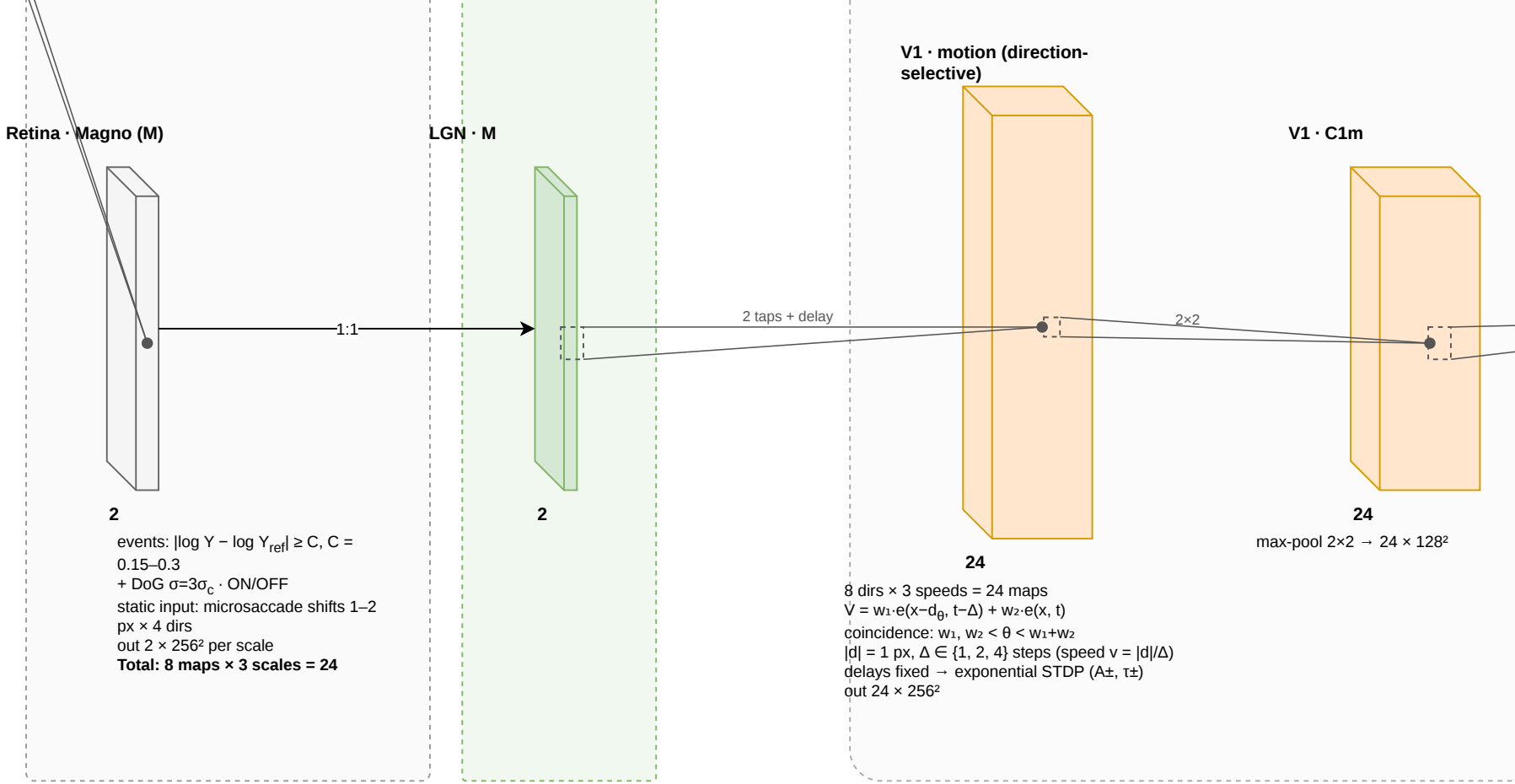
Anatomy of the proposed bio-inspired spiking visual system — wave-based TTFS, ≤ 1 spike per neuron per wave

Each wave: T = 15 time steps per stage · non-leaky IF neurons · lateral inhibition (feature-WTA) in every S layer · wave 1 = pure feed-forward (ultra-rap




pid) · dashed red = feedback applied in wave n+1
ATTENTION FEEDBACK a(x) → all three LGN sheets (applied in wave n+1)
VENTRAL STREAM — what (identity, location-preserving)
V2 thin · S2t · WTA
C2t
5×5
2×2
5×5
16
16 maps, 5×5×12 on C1
W_S2t: 16×12×5×5 = 4,800
surface colour · flat STDP · out 16×128²
16
max-pool 2×2 → 16×64²
V4 · S3 · WTA
V4 · C3
IT · S4 · WTA
2×2
3×3
64
64 maps, 5×5×32 on C2t ⊕ C2p
W_S3: 64×32×5×5 = 51,200
shape × colour · flat STDP
out 64 × 64²
64
max-pool 2×2 → 64 × 32²
128
128 maps, 3×3×64
W_S4: 128×64×3×3 = 73,72
STDP, then R-STDP fine-tu
out 128 × 32²
border-ownership evidence Φ± (wave n+1)
V2 pale · S2p · WTA
C2p
5×5
5×5
2×2
16
16 maps, 5×5×12 on C1
W_S2p: 16×12×5×5 = 4,800
corners, junctions · flat STDP
+ border-ownership B±(x,θ): 8 maps (phase 5)
B± = E · ⌊Φ± − Φ∓⌋+
16
max-pool 2×2 → 16×64²
occlusion (B±)


Selection, goal and memory
Ultra-rapid gist (wave 1)
$t_c^{gist} = \min_u t(D_c(u))$
"is there a car in the scene?"
Attention map → LGN
$a(x) = \mathrm{clip}(\Sigma_k \exp(-\|x-p_k\|^2/2\rho_k^2) - \gamma, -1, 1)$
reward r (R-STDP)
selected p_k
IT · C4
2×2
3×3
128
max-pool 2×2 → 128 × 16²
retinotopic IT map
(no global pooling: location kept)
N_c
Location-wise readout $D_c(u)$
N_c class maps, 3×3×128, shared over u
W_D: N_c×128×3×3
earliest spike per cell wins · silence = background
R-STDP reward per cell: +1 if u∈Ω_c, −1 if u∈$\Omega_{c'}$, $-\lambda_{bg}$ on background
$a_r \propto N_{miss}/N$, $a_p \propto N_{hit}/N$ · dropout
class maps
goal g
K objects
1 (32²)
Priority map (LIP)
$P = \alpha_b \cdot S + \alpha_t \cdot F_g - \alpha_r \cdot R$
S: colour/orientation + motion contrast
F_g: goal-class readout · R: inhibition of return
k-WTA, inhibition radius r → up to K objects
replaces NMS (inside the network)
IT features f_k
PFC · goal + working memory
goal g → F_g (top-down bias)
object files j: p_j, v_j, f_j (IT features), id_j
$\mathrm{cost}(k,j) = \|p_k - (p_j + v_j\Delta t)\| + \lambda(1 - \mathrm{sim}(f_k, f_j))$
unmatched slot: predicted, kept N waves (object permanence)
issues reward r for R-STDP



Legend

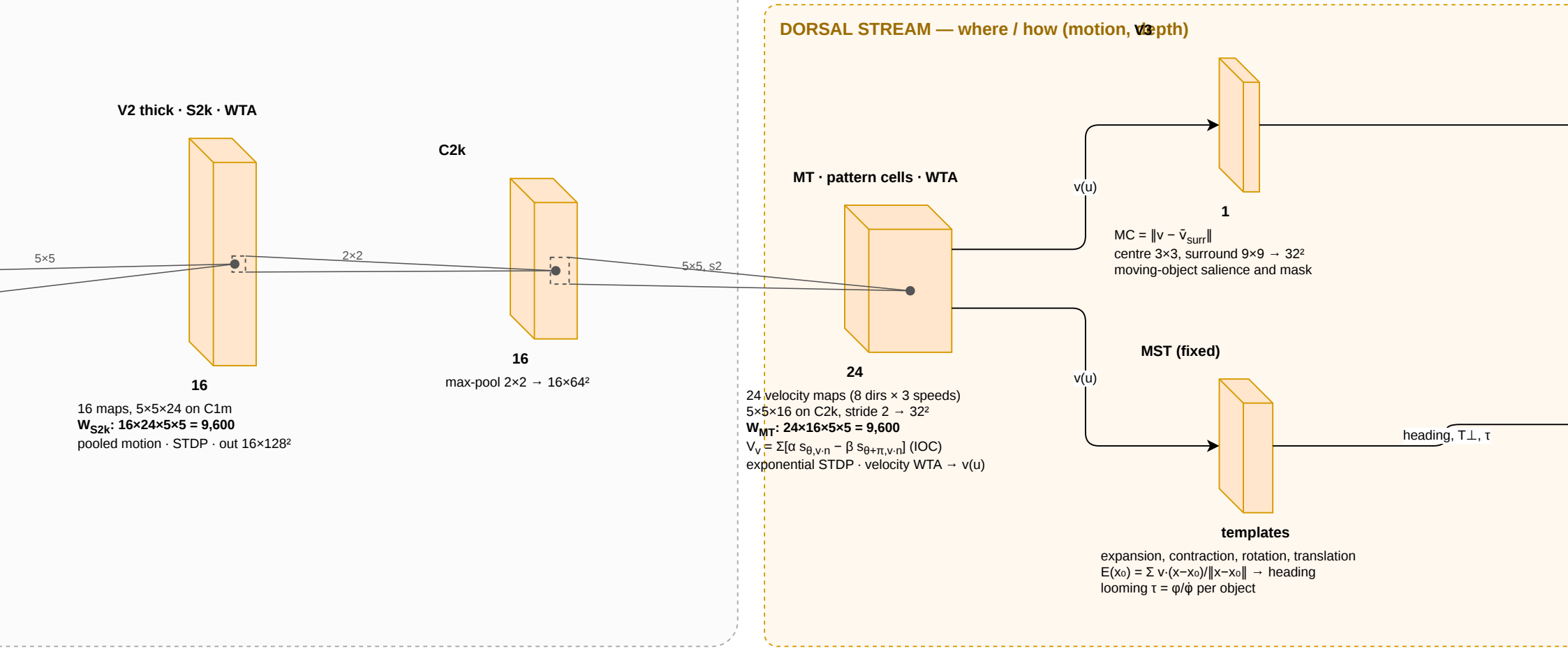
stack of n spiking feature maps (n below the slab)

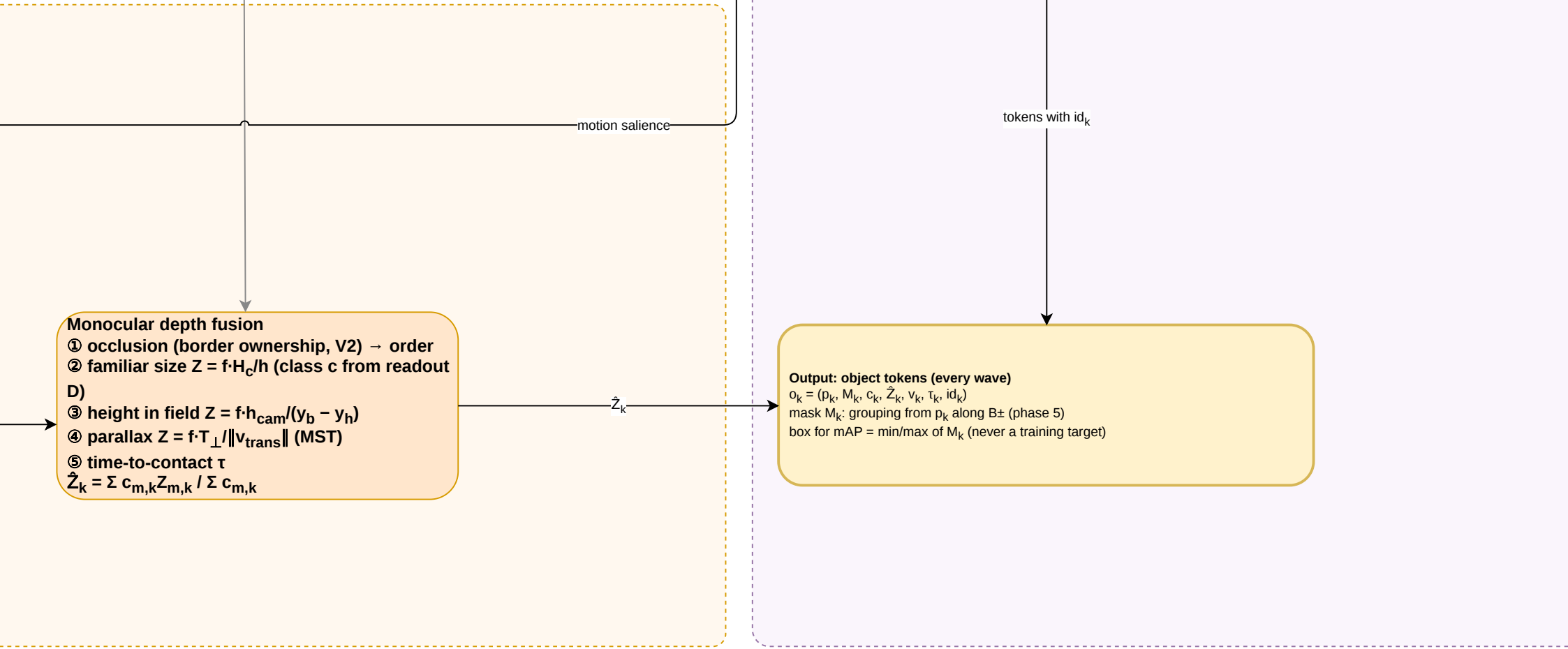
receptive-field window (conv kernel / pooling window)

thin lines = spiking synapses (window → one target neuron)

- - - red dashed = feedback, applied in wave n+1
- - - green dashed = reward signal (R-STDP)
solid arrow = module output · grey = auxiliary input

Colours: grey retina emulator (fixed) · green LGN relay (fixed) · blue ventral / form-colour · orange dorsal / motion · purple selection, attention, PFC · yellow outputs
Learning: flat STDP in S1–S4 and S2*, exponential STDP in motion path, R-STDP in IT fine-tune and readout, fixed MST templates; adaptive threshold $\theta_i \leftarrow \theta_i + \eta_\theta(z_i - \rho)$ in every S layer


V2 thick · S2k · WTA
C2k
5×5
2×2
5×5, s2
16
16
max-pool 2×2 → 16×64²
16 maps, 5×5×24 on C1m
W_S2k: 16×24×5×5 = 9,600
pooled motion · STDP · out 16×128²
DORSAL STREAM — where / how (motion, depth)
V3
MT · pattern cells · WTA
v(u)
1
MC = ‖v − v̄_surr‖
centre 3×3, surround 9×9 → 32²
moving-object salience and mask
24
24 velocity maps (8 dirs × 3 speeds)
5×5×16 on C2k, stride 2 → 32²
W_MT: 24×16×5×5 = 9,600
V_v = Σ[α s_θ,v·n − β s_θ+π,v·n] (IOC)
exponential STDP · velocity WTA → v(u)
v(u)
MST (fixed)
templates
expansion, contraction, rotation, translation
E(x₀) = Σ v·(x−x₀)/‖x−x₀‖ → heading
looming τ = φ/φ̇ per object
heading, T⊥, τ


motion salience
tokens with id_k
Monocular depth fusion
① occlusion (border ownership, V2) → order
② familiar size $Z = f \cdot H_c / h$ (class c from readout D)
③ height in field $Z = f \cdot h_{cam} / (y_b - y_h)$
④ parallax $Z = f \cdot T_\perp / \|v_{trans}\|$ (MST)
⑤ time-to-contact τ
$\hat{Z}_k = \Sigma\, c_{m,k} Z_{m,k} / \Sigma\, c_{m,k}$
$\hat{Z}_k$
Output: object tokens (every wave)
$o_k = (p_k, M_k, c_k, \hat{Z}_k, v_k, \tau_k, id_k)$
mask M_k: grouping from p_k along B± (phase 5)
box for mAP = min/max of M_k (never a training target)